Project Proposal

COMP410 Computer Graphics — Spring 2026

Project Title: Virtual Koç University Campus Tour

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1. Motivation

Koç University has a large and architecturally rich campus that can be difficult to navigate for new students and visitors. The goal of this project is to create an interactive 3D virtual tour of the KU campus exterior. Users will be able to freely walk through an understandable representation of the campus from a first-person perspective. Apart from its practical value for the benefit of visitors, this application can also be regarded as a comprehensive exercise in real-time 3D graphics since it will cover nearly every major topic in COMP410.

2. Project Description

We will implement an interactive first-person 3D campus tour using shader-based OpenGL in C++. The user will navigate through the campus using keyboard and mouse input, observe the buildings and environment under a realistic lighting model, and experience the campus layout as if walking through it in person. We thought it could be better if we split the project into two phases, first of which will cover workload until the progress presentation and the second will cover until the final submission.

Phase 1 - Progress Presentation:

- 3D models of all major academic and social buildings (SCI, ENG, CASE, Law, Library, Sports Center, Student Center)
- Ground plane with basic terrain layout
- First-person camera navigation (WASD + mouse look)
- Perspective projection
- Basic Phong illumination (ambient, diffuse, and specular) with a directional sun light
- Flat-color or basic textured surfaces

Phase 2 - Final Submission:

- Texture mapping on all building facades, ground, and paths
- Skybox (environment cube map)
- Multiple light sources (sun and ambient fill)
- Collision detection so the camera stays on the ground and cannot pass through buildings
- Interactive elements: building labels and information overlays
- Bonus (if time permits): shadow mapping, day/night cycle toggle

3. Computer Graphics Techniques

In this project, we plan to use following techniques, most of which are covered in the course:

Technique	Application in Project
Shader-based OpenGL (GLSL)	All rendering through vertex and fragment shaders
Geometric transformations	Placing and scaling each building in world space
Viewing and Perspective projection	First-person camera using LookAt and Perspective
Phong illumination model	Per-fragment shading on all surfaces

Texture mapping	Building facades, ground, and skybox
Hierarchical / scene graph modeling	Campus scene organized as a tree of objects
Input and interaction	WASD movement, mouse look, keyboard shortcuts
Buffers (VAO/VBO)	Geometry data on the GPU for all objects

4. Technical Approach

Language and API: C++17 with shader-based OpenGL (core profile). We will use GLFW for window and input management, GLEW for extensions, and the Angel.h utilities from the course textbook.

Scene representation: The campus will be organized as a scene graph. The root node represents the campus as a whole, with child nodes for each building. Each node holds its transform (position, rotation, and scale) and a set of mesh components.

Camera system: We will implement a first-person camera defined by a position, yaw, and pitch. The view matrix is recomputed each frame from these values using LookAt(). The camera will be constrained to ground level with basic bounding-box collision against buildings.

Building modeling: Buildings will be constructed from geometric primitives - boxes, cylinders for columns, and flat quads for roofs. Each building is modeled in local space and placed in the world via a model matrix.

Lighting: A single directional light representing the sun will be implemented, with Phong shading computed per-fragment in the GLSL fragment shader.

5. Milestones

Date	Deliverable
March 9, 2026	Proposal submission
Late April 2026	Progress presentation: basic scene, navigation, lighting, at least 3 buildings
End of semester	Final: full campus, textures, skybox, collision detection, and polish

6. Division of Work

Area	Primary
Scene graph and building models	Alperen Kars
Camera system and input	Uğur Öner
Shader development (lighting)	Both
Texture mapping	Both
Collision detection	Alperen Kars
Skybox and environment	Uğur Öner
Report and documentation	Both

Division of work may shift as the project progresses.

7. References

[1] Angel, E. *Interactive Computer Graphics: A Top-Down Approach with OpenGL*, 6th ed., Addison-Wesley.

[2] *OpenGL Programming Guide*, 8th ed., The Khronos OpenGL ARB Working Group.

[3] Koç University Campus Map (official).